# Matthew Embaye

membaye@uwaterloo.ca | (403)-830-5539 | linkedin.com/in/matthewembaye

## Education

**University of Waterloo** **Sep. 2024 – Apr. 2029**
Bachelor of Applied Science, Electrical Engineering Waterloo, ON

- President's Scholarship (2024), Alexander Rutherford Scholarship (2024)

## Experience

**FIRST Robotics, Team 4627** **Nov. 2023 – Mar. 2024**
Electrical Engineering Team Member Calgary, AB

- Built and designed the **"Excellence in Engineering" Award**-Winning robot for Team 4627 in the 2024 Crescendo Challenge using **Autodesk Fusion 360**, wiring, and assembling skills
- Made a robot to pick up and shoot discs into various targets and ultimately hang from a chain being shared with other robots within a specific time while pitted against, and collaborating with, other robots
- Collaborated with a **multidisciplinary ~50-person team** in a communication-focused environment to design, build, and validate chassis, wiring, and hardware prototypes

**Westside Recreation Centre** **Sep. 2023 – Aug. 2024**
Youth Worker Calgary, AB

- Managed queues and supervised volunteers, coordinating new ideas and methods for tasks, while training coworkers and new hires to improve customer service.
- Comfortable with picking things up quickly, being able to reapply and teach it to others, and elevating customer satisfaction through diligence towards communication and customer service skills

**Andaro's Pizza** **Jul. 2021 – Aug. 2021**
Kitchen Staff Calgary, AB

- Exemplified **communication and teamwork** skills by improving store success through prioritization of customer satisfaction in diligence towards quality and care, customer service, and efforts in training others
- Successfully managing the store independently, fulfilling orders, managing disputes, and carrying on various tasks

## Projects

**Arduino Bluetooth Race Car** **Dec. 2023**
Personal Engineering Project Calgary, AB

- Planned, designed, built, and tested a remote-controlled race car utilizing an **Arduino** microcontroller programmed with **C++**, capable of being controlled via mobile device
- Tested and refined various iterations using a variety of technologies including infrared controls, motor controllers, and different chassis utilizing motor and technical skills in circuit design, soldering, and assembly

**Portable Patient Alert System (PPAS) Lead** **Sep. 2024 – Nov. 2024**
Team Lead, Design Studio Project Waterloo, ON

- **Lead a 5-person group** while researching, proposing, designing, prototyping and testing a portable vital reader to check for heart rate and temperature levels, outputting data on an LCD display alongside an alarm system, using a **STM32** microcontroller programmed with **C++**, and various sensors
- Lead designs, delegated tasks, organized deadlines, and lead the process of creating proposals and design documents for council approval, fabricating, testing, and demonstrating prototypes to diligently follow the engineering process

## Technical Skills

**Languages:** C++, C

**Developer Tools:** VS Code, Autodesk Fusion 360, Revit, Altium

**Skills & Interests:** Design, Verification, and Fabrication skills from FIRST robotics, Car, and Vital Reader Projects, Expertise in Computer Literacy (IT), PC building, Troubleshooting Skills, Circuit Design, PCBs, Hardware QA and Testing